

$N=1$ Global Susy in $D=4$:

* Susy algebra is given by $M_{\mu\nu} \oplus P_\mu \oplus Q_\alpha$
s.t.

$$\{Q_\alpha, \bar{Q}^\beta\} = -\frac{1}{2} (\gamma_\mu)_\alpha{}^\beta P^\mu$$

Q_α : 4-component Maj. Spinors

$$\bar{Q} \equiv \lambda^T Q$$

* Particles + anti-particles are identified and transform in irreps of $SO(D-1)$ or $SO(p-2)$

Majorana: $\psi^c = \psi$ (Reality Condition)

$$\psi^c \equiv B^{-1} \psi^*$$

$$[M_{\mu\nu}, Q_\alpha] = -\frac{1}{2} (\gamma_{\mu\nu})_\alpha{}^\beta Q_\beta$$

$$[P_\mu, Q_\alpha] = 0$$

i.e. translations & susy commute.

Dirac Adjoint: $\bar{\psi} = \psi^\dagger \beta$, take $\beta = i\gamma^0$

$$\begin{aligned} \{Q_\alpha, \bar{Q}^\beta\} &= \{Q_\alpha, (Q^\dagger i\gamma^0)^\beta\} \\ &= \{Q_\alpha, i(Q^\dagger)^\delta (\gamma^0)_\delta{}^\beta\} = -\frac{1}{2} (\gamma_\mu)_\alpha{}^\beta P^\mu \end{aligned}$$

$$(\gamma^0)^2 = -1 \quad \{ \gamma^\mu, \gamma^\nu \} = 2\eta^{\mu\nu} 1$$

$$\{Q_\alpha, i(Q^\dagger)^\delta (\gamma^0)_\delta{}^\beta (\gamma^0)_\beta{}^\gamma\} = -\frac{1}{2} (\gamma^0)_\beta{}^\gamma (\gamma_\mu)_\alpha{}^\beta P^\mu$$

$$= -\frac{1}{2} \{Q_\alpha, (Q^\dagger)^\gamma\} = -\frac{1}{2} (\gamma^0 \gamma^\mu)_\alpha{}^\gamma P_\mu$$

$$\therefore \{Q_\alpha, (Q^\dagger)^\gamma\} = \frac{i}{2} (\gamma^0 \gamma^\mu)_\alpha{}^\gamma P_\mu$$

To get the Quantum anti-commutator we simply take the Dirac approach:

$$\{Q_\alpha, (Q^\dagger)^\beta\}_{\text{cl}} \xrightarrow{\text{q}} \{Q_\alpha, Q^{\dagger\beta}\}_{\text{qv}}$$

$$\therefore \{Q_\alpha, (Q^\dagger)^\beta\}_{\text{qv}} = -(\gamma^0 \gamma^\mu)_{\alpha}{}^{\beta} P_\mu$$

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \mathbb{I} \Rightarrow \gamma^0 \gamma^\mu = -\gamma^\mu \gamma^0$$

$$\boxed{\{Q_\alpha, (Q^\dagger)^\beta\}_{\text{qv}} = \frac{1}{2} (\gamma^\mu \gamma^0)_{\alpha}{}^{\beta} P_\mu}$$

* Trace on the spinors:

$$Q Q^\dagger + Q^\dagger Q = \frac{1}{2} (\gamma^\mu \gamma^0)_{\alpha}{}^{\alpha} P_\mu$$

$$\text{Tr}(Q Q^\dagger + Q^\dagger Q) = \frac{1}{2} \text{Tr}(\gamma^\mu \gamma^0) P_\mu$$

$$\Rightarrow \text{Tr}(\gamma^\mu \gamma^0) = -\text{Tr}(\gamma^0 \gamma^\mu)$$

$$\Rightarrow \text{Tr}(\gamma^0 \gamma^\mu) = 0 \quad \forall \mu \neq 0$$

If $\mu=0$, then

$$(\gamma^0)^2 = -\mathbb{I} \quad \text{and so}$$

$$\text{Tr}((\gamma^0)^2) = -4$$

$$\text{Tr}(Q Q^\dagger + Q^\dagger Q) = -2 P_0$$

However $P_0 = -P^0$, hence

$$\boxed{\text{Tr}(Q Q^\dagger + Q^\dagger Q) = 2 P^0}$$

$$\langle \psi | Q_\alpha Q^{\dagger\alpha} + Q^{\dagger\alpha} Q_\alpha | \psi \rangle = 2E$$

Trace over a complete Quantum state yields the total energy of the Hilbert space of a global susy theory.

* R-symmetry: Chiral $U(1)$ w/ generator T_R

$$[T_R, Q_\alpha] = -i (\gamma_\mu)_\alpha^\beta Q_\beta$$

CM-theorem: in the presence of massive particles bosonic charges are limited to M_{max} and P_μ plus (optional) scalar internal symmetry charges. The L.A. is then

$$L.A. \cong SO^\uparrow(1,3) \oplus \underbrace{\mathfrak{g}_{f.c.}}_{\text{finite-compact}}$$

nd Supercharges: Haag-Lopuszanski-Sohnius (HLS) thm: Alg. of symm admits spinor charges Q_α
 • $i=1 \Rightarrow Q_\alpha \rightarrow N=1$ Poincaré susy.

Ex: Majorana

• Momenta $\vec{P}$, $E(\vec{P}) = \sqrt{\vec{P}^2 + m_{B,F}^2}$

Q_α carries angular momenta $3/2$ then

$$\vec{S}|\psi, s\rangle = s|\psi, s\rangle \Rightarrow \vec{S}(\hat{Q}_\alpha|\psi, s\rangle) = (s + 1/2)|\psi, s\rangle$$

$$Q_\alpha|\vec{P}, B\rangle = |\vec{P}, F\rangle \Rightarrow Q_\alpha|\vec{P}, F\rangle \propto |\vec{P}, B'\rangle$$

• Since $[P^\mu, Q_\alpha] = 0$ and so $m_B^2 = m_F^2$ so the states have the same momenta & energy.

(i) Chiral multiplets: $(\chi(x), Z(x), F(x))$

χ : self-conjugate spin $\frac{1}{2}$ field
Majorana

Z : Scalar spin-0 Boson

no χ being self conjugate means

$\chi = \begin{pmatrix} P_L \chi \\ P_R \chi \end{pmatrix}$, $\chi \rightarrow P_2 \chi$ and
we can decompose $Z(\chi)$ as

$$Z(\chi) = (A(\chi) + i B(\chi)) / \sqrt{2}, \quad A(\chi), B(\chi) \in \mathbb{R}$$

(ii) Gauge multiplet: $(A_\mu(\chi), \lambda(\chi))$

$SO(1, 3)$: Massless multiplet

$$P_\mu = (E, 0, 0, E) \Rightarrow SO(2)$$

$$\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \simeq \begin{array}{|c|} \hline \\ \hline \end{array} \text{ of } U(1)$$

$$\frac{4 \cdot 3}{2} \simeq 6?$$

• Is the adjoint rep of $SO(2)$ 6?
Don't think so...

• $\lambda(\chi)$: Gaugino, spin $\frac{1}{2}$

• $Q_\alpha = \int d^3x \mathcal{T}_\alpha^0(\vec{x}, t) \mid \mathcal{T}_\alpha^\mu$: supercurrent

(i) Consider the free scalar field $\phi(x)$

$(\square - m^2) \phi = 0$ and a Dirac field $\Psi(x)$ satisfying

$$(\not{\partial} - m) \Psi = 0$$

ex 6.2] $T^{\mu} = (\partial\phi - m\phi)\gamma^{\mu}\psi$

proof:

$$\begin{aligned} \partial_{\mu} T^{\mu} &= \partial_{\mu} (\partial\phi - m\phi)\gamma^{\mu}\psi + (\partial\phi - m\phi)\gamma^{\mu}\partial_{\mu}\psi \\ &= \partial_{\mu} (\partial\phi - m\phi)\gamma^{\mu}\psi + (\partial\phi - m\phi)\gamma^{\mu}\partial_{\mu}\psi \\ &= (\gamma^{\mu}\partial_{\mu}\partial\phi - m\partial_{\mu}\phi)\gamma^{\mu}\psi + m(\partial\phi - m\phi)\psi \\ &= (\gamma^{\mu}\partial_{\mu}\partial\phi - m\partial_{\mu}\phi)\gamma^{\mu}\psi + (m\partial\phi - \square\phi)\psi \\ &= \gamma^{\mu}\gamma^{\mu}\partial_{\mu}\partial\phi - \square\phi\psi \end{aligned}$$

okay...

* Note that the term $T^{\mu}_{\mu} = (\partial\phi - m\phi)\gamma^{\mu}\partial_{\mu}\psi$ violates the HLS theorem. However it is conserved.
 $\Rightarrow$ Cannot be conserved in an interacting theory

* Interactions: $\phi \rightarrow Z(x)$
 $\psi_D \rightarrow \psi_M$

• Gauge multiplet with vector potential A_{μ} and $F_{\mu\nu} = \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}$ satisfying Maxwells eqs
 $\partial^{\mu}F_{\mu} = 0$, free spin-1 field $\partial^2\lambda = 0$

$$T^{\mu} = \gamma^{\mu\rho} F_{\rho\sigma} \gamma^{\sigma\mu} \lambda$$

is also conserved.

$$\partial_{\mu} T^{\mu} = \partial_{\mu} F_{\rho\sigma} \gamma^{\mu\rho} \gamma^{\sigma\mu} \lambda + \gamma^{\mu\rho} F_{\rho\sigma} \partial_{\mu} \gamma^{\sigma\mu} \lambda$$

use:

$$\gamma^{\mu\rho} \gamma^{\sigma\mu} = \gamma^{\mu\rho\sigma} + 2\gamma^{\mu\rho}\gamma^{\sigma}$$

$$\partial_{\mu} F_{\rho\sigma} (\gamma^{\mu\rho\sigma} + 2\gamma^{\mu\rho}\gamma^{\sigma}) \lambda$$

Bianchi

Maxwell